MECHATRONICS / ARTIFICIAL INTELLIGENCE / ROBOTICS

Felipe Rojas



Mechatronics Engineer, M.Eng.
Robotics, software and intelligent systems

Ingolstadt, Germany · Open to relocation and travel

+49 176 2514 0535 · feliperojas.ec@gmail.com

Web portfolio / GitHub / LinkedIn

Mechatronics engineer with a master's degree in artificial intelligence and experience in robotics, research software and the automotive industry. I combine Python and C++ development with sensor integration, simulation, functional testing and structured quality management.

TECHNICAL PROFILE

Robotics & sensors

UR3 · mobile robots
ROS / ROS 2 · sensor fusion
Robotic sensing

Software & testing

Python · C / C++
Git / GitHub · Linux
Docker · pytest · CI

Simulation & integration

Gazebo · RViz · URDF
SolidWorks · MATLAB
Functional and integration tests

AI & computer vision

OpenCV · deep learning
I3D · LSTM · MLP
Detection and tracking

Automation & quality

Siemens S7 · TIA Portal
(academic practice)
ISO 9001 · IATF 16949
Technical documentation

Available immediately

RECENT TECHNICAL PROJECTS

Extended Kalman Filter — 10/2026

Modernized Udacity course project · GitHub

C++17 sensor fusion using an Extended Kalman Filter, with automated tests and CI.

ROS 2 Multimodal Grasp Monitor — 09/2026 - 10/2026

Independent simulation project · GitHub

Simulated proximity, force and vibration signals to monitor grasp states; rule-based evaluation and tests.

Finding Lane Lines — 09/2026

Modernized Udacity course project · GitHub

Camera-based lane detection in Python/OpenCV and C++17/OpenCV.

Automotive CAN Validation Lab — 07/2026 - 08/2026

Independent Python learning project · GitHub

Data model for classical CAN frames with validation logic, pytest and CI.

TECHNICAL WORK EXPERIENCE

Student Research Assistant — 02/2023 - 09/2024

Electronics and Robotics Laboratory
TH Deggendorf, Cham Campus

- Set up Gazebo environments for QCar and UR3 using ROS 1, C++ and Python for research and student projects; configured TurtleBot networking and remote access.
- Developed a Python Rosbag reader for ModuSLAM to identify data types and convert them to the mapping pipeline's input format; configured with Hydra.
- Extended pytest coverage to additional data types within GitHub CI/CD workflows.

Experience & education

LANGUAGES

German

C1 · self-assessed
telc Deutsch B2 · 07/2026
C1 intensive course · 09/2026

English

C2 · self-assessed
TOEFL iBT: 108/120 · 2022

Spanish

Native language

Other languages

Italian: A1, self-assessed
Japanese: JLPT N5

FURTHER TRAINING

OpenCV & Deep Learning

Udemy · 09/2026
Course certificate awarded

Self-Driving Car Engineer

Udacity Nanodegree
07/2020 - 01/2021

SOLIDWORKS CSWA

Certification · 06/2014

PUBLICATION

Case Study: Autonomous Mobile Robot Exploration and Navigation in Unknown Maze Environment

IEEE CoDIT 2024
Co-author

Read on IEEE Xplore

TECHNICAL EXPERIENCE · CONTINUED

Production Engineer

08/2020 - 08/2022

Tecnividrio 2000 S.A., Ecuador · Automotive Tier 1

- Built an Excel-based production and quality documentation system largely from scratch; used Python to identify bottlenecks and defect-prone process stages.
- Aligned material inventory and production with monthly delivery needs during COVID-related supply shortages to avoid overproduction.
- Supported ISO 9001 and IATF 16949 audit preparation; supervised product changeovers and trials on the Keraglass furnace as needed.

Research & Development Intern

08/2017 - 10/2017

Mundy Home, Ecuador

- Developed a LabVIEW GUI for quality data and supported screw-tightening verification tests in radio assembly.

Student Technical Assistant

05/2017 - 07/2017
02/2016 - 07/2016

Tecnividrio 2000 S.A., Ecuador

- Created automotive-glass setup manuals and assisted with glass-tempering furnace installation.

ADDITIONAL EMPLOYMENT

Event Catering (Part-time)

08/2025 - present

Stadium operations, Nuremberg

Warehouse Logistics

08/2025 - 04/2026

Geis, Erlangen · Siemens Healthineers site

Food Service (During Studies)

07/2023 - 09/2023

Cham, Germany

EDUCATION

M.Eng. Artificial Intelligence for Smart Sensors and Actuators

10/2022 - 06/2025

TH Deggendorf · Graduated: 26 June 2025 · Grade: 1.7 (Germany)

Robotics and autonomous systems, sensors and actuators, model-based function development, verification and validation.

Master's thesis: Intention Recognition of Vulnerable Road Users in the Frame of Human-Centered Decision Making for Autonomous Driving

B.Eng. Mechatronics

07/2013 - 12/2018

UTE University, Quito, Ecuador

Robotics, PLCs and industrial communication, control engineering and mechatronic system integration.